$\Sigma = \{0, 1\}$

$(a|b)^*$

$\{\phi\}$

$\{\varepsilon\}$

Lexical Analysis ≡ Scanning

$0^* = \{\varepsilon, 0, 00, 000, \dots\}$

regular expression

$0^+ = \{0, 00, 000, \dots\}$

$0^1^* = \{0, 01, 011, 0011, \dots\}$

$0^+1^* = \{0, 01, 011, 0011, \dots\}$

$0|1 = \{0, 1\}$

$0^*1^* = \{\varepsilon, 0, 1, 01, 0011, 001, 000, 111111 \dots\}$

0ε $\varepsilon 1$

$(01)^* = \{\varepsilon, 01, 0101, 010101, \dots\}$

$0^*1^* \neq (01)^*$

$(a|b)^+ = \{a, b, aaa, bbb, ababbb, \dots\}$

$0^*1|1^+0 =$

$\{1, 01, 001, 0001, 10, 110, 1110, \dots\}$

$a^+b^*|c^+d = \{a, ab, abb, \dots, cd, ccd, cccd, \dots\}$

$a^*|b^* = \{\varepsilon, a, aa, aaa, \dots, b, bb, bbb, \dots\}$

$(a|b)^* = \{\varepsilon, a, aaa, b, bbb, ab, ba, abbbaa, baabbb, \dots\}$

$a^*|b^* \neq (a|b)^*$

Consider the alphabet $\Sigma = \{a, b\}$

Construct the regular expressions of the following.

a) produce strings with at least two a's.

$(a|b)^* a (a|b)^* a (a|b)^*$

b) produce strings with exactly two a's.

$b^* a b^* a b^*$

$\Sigma = \{a, b, c\}$

e)

$(b|c)^* a a (b|c)^*$

e) produce strings with exactly two consecutive a's

$b^* a a b^*$

Lexical Analysis $\equiv$ Scanning

c) produce strings with at least one a followed by at least one b.

$(a|b)^* a (a|b)^* b (a|b)^*$

d) produce all possible strings of a's, and b's.

$(a|b)^*$

$\{a, b, c\}$

$(a|b)^* = \{a, b, aae, bbb, abababbb, \ldots\}$

$ab^*|c^*d = \{a, ab, abb, \ldots, cd, ccd, cccd, \ldots\}$

baccacba

at least one a followed by exactly two consecutive b's

$b^* (a^+ bb)^+$

$a^* a = a^+$